

Italy:

The state of High Temperature Superconductivity in 2025

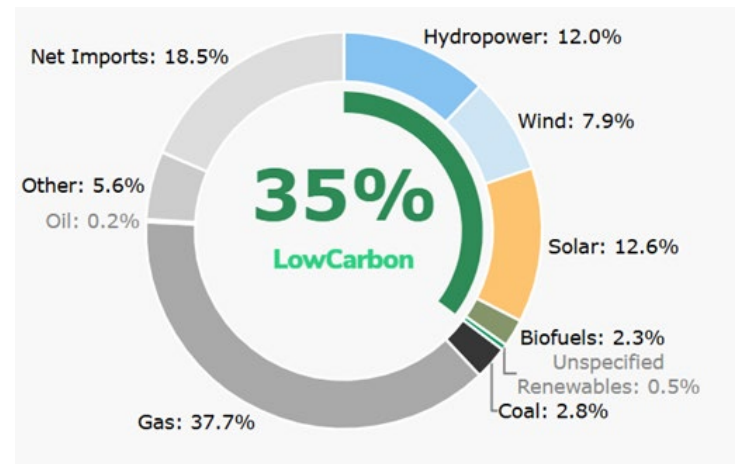
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Energy Context

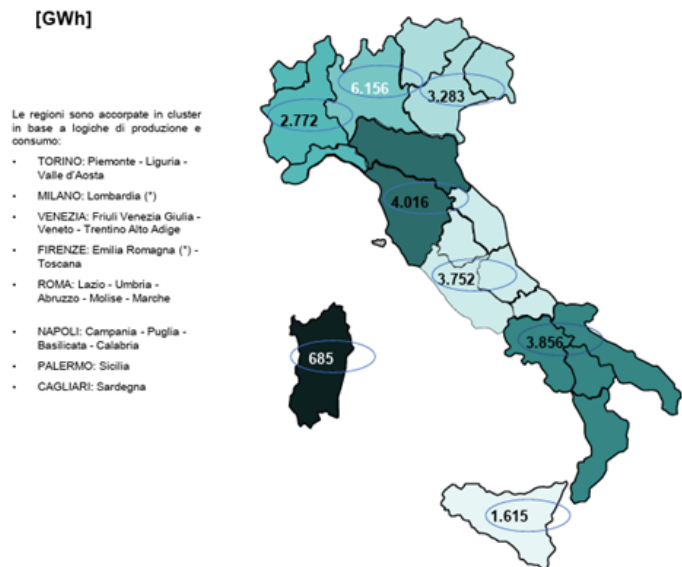
In 2025, Italy's power system is characterized by a combination of fossil fuels and low-carbon sources. According with the data reported in Figure 1, fossil energies still play a major role, making up roughly 41% of electricity use, with gas accounting for nearly 38% and coal contributing a small 3%. Low-carbon sources provide about 35% of the overall mix: solar leads with around 13%, followed closely by hydropower at 12%, while wind supplies about 8%. Net electricity imports—which include both clean and fossil-based generation—represent approximately 19%. Biofuels, although considered low-carbon, make only a modest contribution of about 2%.

According to the Italian Transmission System Operator Terna the energy demand totalised 311,3 TWh, the most coming from Northern Regions. On the contrary, wind and solar resource are higher in the South than in the North of the country. This is confirmed by the geographical distribution of Photovoltaic and wind farms already operating, and in the pipeline. PV requests totalise 145.98 GW, onshore wind 70.21 GW, and offshore wind 107.59 GW.



[Italy power system shares by sources](#)

Fabbisogno suddiviso per Aree Territoriali – Rappresentazione territoriale



Fonte: Terna

Share of energy demand in different areas of Italy (period December 2025)

Policy and Regulatory Highlights

Italian energy policies are heavily influenced by European targets (Fit for 55, REPowerEU) and primarily focus on decarbonization, aiming to accelerate the transition towards a low-carbon energy system, and energy security aiming to reduce the strong dependence on natural gas imports, especially following the last years geopolitical crisis.

At national level, the National Energy and Climate Plan (NECP) is the main strategic document defining targets and measures up to 2030. In particular a strong growth of PV and Wind capacity. In particular a strong growth of PV and Wind capacity is expected to reach the ambitious 2030 target of 63.4 % of renewable in the electricity consumption.

The National Recovery and Resilience Plan is another Pandemic European measure that provides significant financial resources for investments in the green transition, including renewables, grids, green hydrogen, and efficiency. This plan is going to conclude in 2026. In this frame a big project, IRIS (see in the following) on superconductivity has been funded.

Since 2023 a debate has reopened on the possibility of reintroducing nuclear energy (new generation, such as small modular reactors - SMRs) into the future energy mix, although there are currently no concrete operational decisions.

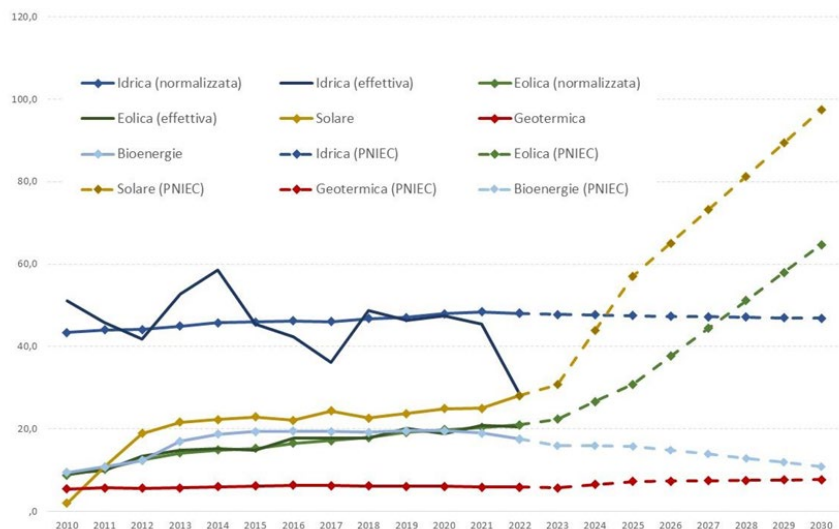
Infrastructure Highlights

TERNA plans major electricity grid reinforcements, with over €23 billion of investments by 2034, including new HVDC interconnections such as the Tyrrhenian and Adriatic Links, increasing cross-zone transmission capacity by about 7 GW by 2030 and enabling the integration of more than 65 GW of new renewable generation.

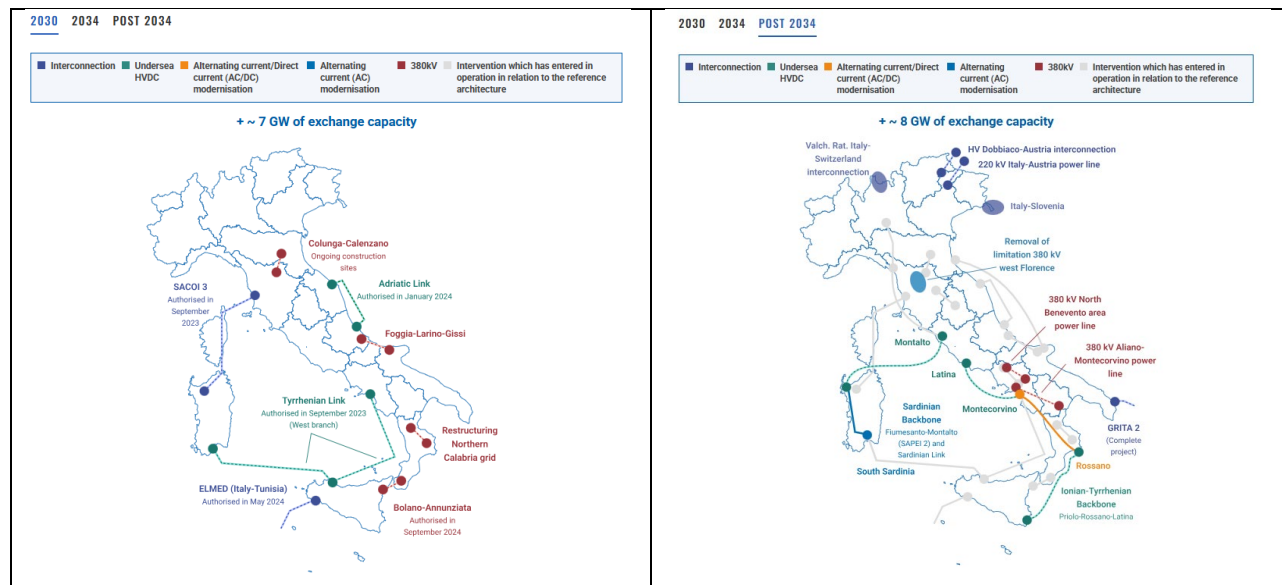
Strategic infrastructure will be set in place to increase intrazonal security and capacity. Such interventions will promote the energy exchange within a market zone, in order to integrate renewable sources and solve any grid congestion issues.



Regional share of the Request of connection received by Terna as of 31st January 2026. (Orange: PV; Green: onshore wind; Blue: offshore wind).



2010-2030 trend of renewable production growth (TWh by source) to reach NECP targets (source [MASE](#) – Elaborated by GSE and RSE.)



TERNA intrazonal interventions by 2030 (left) and post 2034 (right)

The 2025 Development Plan was presented in Rome on 14 March by Giuseppina Di Foggia - Chief Executive Officer and General Manager of Terna – and includes also a strong reinforcement of the interconnections between Italy and Europe, North Africa and the Middle East.

R&D Activities and Demonstration Projects Highlights

Many projects were active in 2025 in Italy and the main are described below.

R&D Activities

Ricerca di sistema (RdS): it is a national three years research programme (2025-2027) on many topics concerning electric and energetic system. Activities on superconductivity applications in the power grid total more than 1 M\$/y. The main topics concern the thoric and experimental study of AC and DC SFCL and the techno-economic assessment of SC cables. RSE is the coordinator and the main beneficiary and collaborates with University of Bologna, Politecnico of Turin and University of Padoa.

PNRR H2: it is a project funded in the frame of the National Recovery and Resilience Plan ending in 2026. In this project RSE is carrying out simulation and test on SC cables cooled by liquid hydrogen.

Demonstration Projects

EU project SCARLET (“Superconducting cables for sustainable energy transition”) aims at developing and industrially manufacturing superconducting cable systems at the gigawatt level, bringing them to the last qualification step before a commercial installation. The case study SCARLET focuses on consists in the possibility to connect renewable energy sources (RES) to the power grid by superconducting cables, even when RES are remotely located both onshore and offshore. Lasting from September 2022 until February 2027, Scarlet involves 15 partners from 7 different countries under the coordination of SINTEF (Norway). Basically, three types of demonstrators are foreseen: $\pm 50\text{kV}/10\text{kA}$ onshore/offshore MVDC HTS cable demonstrator in liquid nitrogen (70 K), $\pm 25\text{kV}/20\text{kA}$ onshore MVDC MgB₂ cable demonstrator in liquid hydrogen (20 K), system protection with 50kV/10kA DC superconducting fault current limiter module demonstrator. The project is articulated in 7 Work Packages (WPs), including both technical WPs for the development of the demonstrators and the management/disseminations activities. SCARLET will investigate the whole range of effects of superconducting cables along with their protection and cryogenic systems, as their properties open the door for cost savings beyond the cables themselves; in addition It offers a unique possibility to study the possibility to transfer energy through two different energy vectors (hydrogen and electricity) with the same infrastructure.

The IRIS project, funded under the Italian National Recovery and Resilience Plan (PNRR) within the post-COVID “Recovery Funds” supported by the EU Next Generation Europe programme, aims to establish a new distributed research infrastructure for applied superconductivity in Italy. The initiative involves six poles across the country, combining and expanding existing infrastructures, competences, and laboratories, with LASA in Milano acting as the main hub and coordinating the activities. Launched on 1 November 2022, the project runs until October 2025 and addresses multiple aspects of superconductivity, with a strong focus on technological development and experimental validation. Among its key activities is the design, development, and validation of advanced superconducting prototypes, including a 10 T HTS dipole and a high-power transmission line based on superconducting technology. In particular, a 1 GW (40 kA – 25 kV) 130 m long cable based on MgB_2 and refrigerated at 20 K is being developed along the full value chain, encompassing cable architecture, conductor integration, cryogenic compatibility, and operational reliability. The objective is to demonstrate the technical feasibility and system-level benefits of superconducting transmission for future resilient and sustainable power grids. The project also includes the construction of a dedicated test facility at the University of Salerno, designed for the experimental validation of superconducting cables and associated components under realistic electrical, thermal, and operational conditions, such as cryogenic operation, high current loading, and transient scenarios. This infrastructure will enable performance assessment, safety verification, and long-term reliability testing, supporting both research activities and technology transfer, and representing a strategic asset for advancing superconducting technologies and reducing the risks associated with their deployment in real power system applications.

Industrial Players and Activities Highlights

A significant number of Italian players are active in the field of superconductivity applications in the power grids.

ASG Superconductors: The Italian company ASG Superconductors has developed manufacturing processes enabling the industrial production of MgB_2 -based conductors. The company’s approach is based on established metallurgical techniques, resulting in composite wires in which MgB_2 powder is embedded within a metallic matrix. This configuration enables high current-carrying capacity compared to conventional conductors such as copper, while maintaining mechanical robustness and manufacturability. The decision to keep production and know-how entirely in Italy is not just a matter of industrial pride: it ensures control over the value chain, supply security, and the ability to respond quickly to the needs of major international clients. An important characteristic of MgB_2 -based systems is their operating temperature, typically around 20 K. This allows the use of commercially available cryogenic technologies, including those developed for liquefied natural gas (LNG) applications. As a result, integration into existing or planned infrastructure may be facilitated, potentially reducing deployment costs compared to superconductors requiring lower operating temperatures. Recent developments indicate increasing technological maturity. More than 2.500 kilometres of MgB_2 superconducting wire have been produced for applications including power cables, magnets and grid support systems. Moreover, it is a well-established technology that has already accumulated millions of operating hours in various systems produced by ASG in collaboration with partners, mainly research centres. Actually in ASG’s laboratories, within the IRIS project—funded by public programs as Italy’s National Recovery and Resilience Plan (PNRR)—the most powerful energy transmission cable ever built in Europe (1 GW) is under construction.



One example of ASG 25kV/1GW MgB_2 cable

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ICAS: ICAS S.r.l. (Innovation and Consulting on Applied Superconductivity), is a private company originally setup in 2010 as a Consortium (including Tratos Cavi and Criotec Impinati), in order to carry out the big duty of

producing superconducting conductors for the ITER and the JT-60SA tokamak experimental reactors. In the following years ICAS was awarded supply contracts from all over the world and the range of production activities grew up consequently. Currently, ICAS recognized excellence is not limited to superconducting Cable-In-Conduit conductors manufacturing, but covers also other and diversified high technology productions and services.

Suprema: Suprema Tape S.r.l. is a deep-tech startup based in Rome developing and manufacturing High Temperature Superconductor (HTS) tapes using Pulsed Laser Deposition (PLD) technology. The company was founded by four physicists from ENEA, Italy's national agency for new technologies, energy and sustainable economic development, who bring decades of combined research expertise directly into an industrial manufacturing context. Suprema's primary markets are fusion energy and grid infrastructure, two sectors where demand for high-performance HTS tapes is growing rapidly and where supply remains critically constrained. Additional high-value applications include electric mobility, NMR systems, particle accelerators, and industrial superconducting systems. To date, Suprema has completed a full industrial-scale validation of its production process, demonstrating that its tapes consistently achieve performance levels that set a new benchmark in the market. The results confirm the technical maturity of Suprema's approach and its readiness to transition to commercial manufacturing.

Alongside this milestone, the company has built direct relationships with leading customers across all its target markets and has developed the proprietary advanced production systems that will form the backbone of its first facility. Suprema is now closing a seed funding round that will finance the construction of its first production facility in the Rome area. Commercial production is targeted for early 2028.

Success Stories and Lessons Learned

2025 Success Story: The Suprema Start-Up:

“Suprema represents a fundamental step for the growth of the superconductivity sector in Europe and helps to rebalance the production gap with Asia, where 90% of global production is currently concentrated, by promoting the continent’s technological self-sufficiency and becoming a reference point for it,” said researcher and Suprema CEO Andrea Augieri, who founded the startup together with Fabio Fabbri, Francesco Rizzo, and Giuseppe Celentano, also researchers at ENEA’s Nuclear Department ([translated](#)).

